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| PROFILE                   | <p><b>Affiliation:</b> University of Oxford</p> <p><b>Email:</b> richard.stiskalek@physics.ox.ac.uk</p> <p><b>NASA ADS:</b> 19 first-author, 17 co-author papers; <i>h</i>-index = 13; citations = 901</p> <p><b>Website:</b> richard-sti.github.io</p> <p>I am a Career Development Research Fellow at St John’s College, Oxford, and a Hintze Prize Fellow in Astrophysics (Cosmology) at the University of Oxford. I study how structure formed in the nearby Universe, out to a few hundred megaparsecs, and how fast it is expanding. I specialise in Bayesian field-level inference with constrained simulations—digital twins of the local Universe—combined with semi-analytic galaxy formation models and machine learning to test galaxy physics and map large-scale flows. I implement these as GPU-accelerated, differentiable models in Python and JAX, sampling millions of parameters on high-performance computing clusters.</p>                                         |
| EDUCATION                 | <p><b>University of Oxford</b> (Balliol College), DPhil in Astrophysics 2022 – 2026<br/>Thesis: “Constrained Simulations of the Local Universe as a Laboratory for Precision Cosmology and Astrophysics”<br/>Supervised by <i>Harry Desmond</i>, <i>Julien Devriendt</i>, and <i>Adrianne Slyz</i></p> <p><b>Ludwig-Maximilians-Universität München</b>, MSc in Physics 2020 – 2022<br/>Thesis: “Strong-field lensing of gravitational waves” (partially with <i>Albert Einstein Institute Potsdam</i>)<br/>Supervised by <i>Miguel Zumalacárregui</i>, <i>Marius A. Oancea</i>, and <i>Jochen Weller</i></p> <p><b>Hong Kong University of Science and Technology</b> 2017 – 2018<br/>Undergraduate Exchange Programme in Physics</p> <p><b>University of Glasgow</b>, BSc in Physics with Astrophysics 2016 – 2020<br/>Thesis: “Gravitational-wave cosmology”<br/>Supervised by <i>Martin Hendry</i></p>                                                                               |
| ACADEMIC APPOINTMENTS     | <p><b>University of Oxford</b>, St John’s College 2026 – present<br/>Career Development Research Fellow</p> <p><b>University of Oxford</b>, Hintze Centre for Astrophysical Surveys 2026 – 2029<br/>Hintze Prize Fellow in Astrophysics (Cosmology)</p> <p><b>Flatiron Institute</b>, Center for Computational Astrophysics 2025<br/>Pre-Doctoral Fellow (with <i>Shy Genel</i> and <i>Lucia A. Perez</i>)</p> <p><b>Institut d’astrophysique de Paris</b> 2024<br/>Long Term Attachment (with <i>Guilhem Lavaux</i>)</p> <p><b>Max Planck Institute for Gravitational Physics</b>, Observational Relativity and Cosmology 2020<br/>Summer Research Intern (with <i>Collin Capano</i>)</p> <p><b>University of Oxford</b>, Sub-department of Astrophysics 2019<br/>Summer Research Intern (with <i>Harry Desmond</i>)</p> <p><b>University of Glasgow</b>, Institute for Gravitational Research 2018<br/>Summer Research Intern (with <i>John Veitch</i> and <i>Chris Messenger</i>)</p> |
| COLLABORATION MEMBERSHIPS | <p><b>Learning the Universe</b> 2025 – present<br/>Simons Foundation collaboration developing next-generation cosmological simulations, machine learning methods, and advanced Bayesian inference for large-scale surveys<br/><i>Contributions:</i> Training set generation and BORG team</p> <p><b>CAMELS</b> 2025 – present<br/>Collaboration building large public suites of cosmological hydrodynamical and <i>N</i>-body simulations spanning broad cosmological and astrophysical parameter ranges to train and test machine learning methods<br/><i>Contributions:</i> Leading projects on semi-analytic modelling and simulation-based inference</p>                                                                                                                                                                                                                                                                                                                             |

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|                                  | <b>Aquila Consortium</b><br>International collaboration developing Bayesian field-level inference techniques to reconstruct initial conditions and study large-scale structure formation<br><i>Contributions:</i> Leading projects on data generation, benchmarking, large-scale flows and cosmological parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2022 – present                                                                                                                           |
| TEACHING, MENTORING & SERVICE    | <b>Student Supervision</b><br><i>Noah Pierce</i> (BSc, Southampton; with H. Desmond & T. Yasin) “Field-Level Inference of the Galaxy Luminosity Function”<br><i>Joshua Darne</i> (MPhys, Oxford; with T. Yasin & H. Desmond) “Radial acceleration relation in the NewHorizon hydrodynamical simulation”<br><i>Fedir Boreiko</i> (BSc, Manchester; with T. Yasin & H. Desmond) “The correlation between light and dark matter across cosmic time” [arXiv:2601.07799]<br><i>Enoch Ko</i> (BSc, Warwick; with T. Yasin & H. Desmond) “Dark matter and galaxy dynamics: enduring puzzles” [arXiv:2508.03569]<br><i>Catherine Gallagher</i> (MPhys, Oxford; with T. Yasin & H. Desmond) “The influence of cosmic environment on galaxy properties” [arXiv:2503.14732]<br><i>James Harvey</i> (BSc, Oxford; with T. Yasin & H. Desmond) “Machine learning the time of last major merger from spectroscopic data” | 2025 – present<br>2024<br>2024<br>2024<br>2023 – 2024<br>2023 – 2024                                                                     |
|                                  | <b>Teaching Experience</b><br>Tutor, <i>MPhys C1 Astrophysics</i> , University of Oxford<br>Graduate tutorials in cosmology and stellar astrophysics<br>Demonstrator, <i>3rd-Year Practical Course</i> , University of Oxford<br>Computational astrophysics practical sessions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2023 – 2024<br>2023                                                                                                                      |
|                                  | <b>Academic Service</b><br>Referee for <i>A&amp;A</i> , <i>ApJ</i> , <i>MNRAS</i> , <i>OJA</i> , <i>PNAS</i> , and <i>PRD</i><br>Reviewed 20+ manuscripts for leading journals<br>Coordinator, Aquila Consortium Monthly Telecon<br>Organiser, “Simulator’s Coffee”, University of Oxford<br>Local organiser, Aquila Consortium Oxford Meeting<br>Lead organiser, “Middle of Scotland Science Festival”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022 – present<br>2023 – present<br>2023 – 2024<br>2023<br>2018                                                                          |
| SELECTED AWARDS AND SCHOLARSHIPS | Hintze Prize Fellowship in Astrophysics (Cosmology), University of Oxford<br>Career Development Research Fellowship, St John’s College, Oxford<br>Flatiron Research Fellowship, Flatiron Institute (declined)<br>Junior Research Fellowship, Emmanuel College, Cambridge (declined)<br>Snell Exhibition, Balliol College<br>STFC PhD Funding, Science and Technology Facilities Council<br>DAAD Study Scholarship, German Academic Exchange Service<br>Kerr Bursary, University of Glasgow<br>Lang Scholarship, University of Glasgow<br>Undergraduate Summer Bursary, Royal Astronomical Society<br>Dean’s List, Hong Kong University of Science and Technology<br>Astronomy 1 Prize, University of Glasgow<br>Matthew A. Muir Bursary, University of Glasgow<br>South East Asia Study Abroad Scholarship, University of Glasgow                                                                          | 2026<br>2026<br>2026<br>2026<br>2022 – 2026<br>2022 – 2026<br>2021 – 2022<br>2020<br>2019<br>2018<br>2018<br>2017<br>2017<br>2017 – 2018 |
| SKILLS                           | <b>Programming:</b> Python, Julia, C/C++, Fortran; high-performance computing (MPI, GPU acceleration)<br><b>Statistics &amp; Machine Learning:</b> Bayesian inference (NumPyro/JAX, PyMC, Stan), simulation-based inference, density estimation; deep learning (MLPs, CNNs, GNNs, normalising flows) with JAX, TensorFlow, PyTorch; tree-based methods (random forests, gradient boosting) with scikit-learn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                          |

**Astrophysics Software:** RAMSES, AREPO, Gadget, Rockstar, DisPerSe; *N*-body/hydrodynamical simulation analysis

**Languages:** English (fluent), Czech (native), Slovak (fluent), French (intermediate), German (beginner)

## PUBLICATIONS

### First-Author – Submitted

- [1] “Two-rung ladder:  $H_0$  from Tip of the Red Giant Branch and geometric anchors alone”; **R. Stiskalek**, H. Desmond, G. Lavaux [arXiv:2609.29996]
- [2] “A reanalysis of the megamaser Hubble constant: from spot catalogues to peculiar velocities”; **R. Stiskalek**, H. Desmond [arXiv:2609.17684]

### First-Author – Accepted

- [3] “Learning the Universe with cosmological rescaling of merger trees and semi-analytic galaxy formation models”; **R. Stiskalek**, L. A. Perez, S. Genel, R. S. Somerville, R. E. Angulo, S. Contreras [ApJ 1007:112, arXiv:2606.10024]
- [4] “Forward-modelling Milky Way Cepheids: selection effects and physical priors in the Gaia–HST calibration”; **R. Stiskalek**, A. Riess, H. Desmond, G. Lavaux, D. Scolnic [MNRAS 550:stag1272, arXiv:2603.09880]
- [5] “ $S_8$  from peculiar velocities: agreement with *Planck* for Tully–Fisher and supernovae, tension for the fundamental plane”; **R. Stiskalek** [MNRAS 550:stag1266, arXiv:2509.20235]
- [6] “Validating Digital Twins of the Local Universe with the Thermal Sunyaev-Zel’dovich Signal”; **R. Stiskalek**, H. Desmond [Open Journal of Astrophysics 9, arXiv:2601.15935]
- [7] “Revisiting the Great Attractor: The Local Group’s streamline trajectory, cosmic velocity and dynamical fate”; **R. Stiskalek**, H. Desmond, S. McAlpine, G. Lavaux, J. Jasche, M. J. Hudson [Open Journal of Astrophysics 9, arXiv:2601.08524]
- [8] “1.8 per cent measurement of  $H_0$  from Cepheids alone”; **R. Stiskalek**, H. Desmond, E. Tsaprazi, A. Heavens, G. Lavaux, S. McAlpine, J. Jasche [MNRAS 546:staf2260, arXiv:2509.09665]
- [9] “No evidence for local  $H_0$  anisotropy from Tully–Fisher or supernova distances”; **R. Stiskalek**, H. Desmond, G. Lavaux [MNRAS 546:staf2048, arXiv:2509.14997]
- [10] “The Velocity Field Olympics: Assessing velocity field reconstructions with direct distance tracers”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz, G. Lavaux, M. J. Hudson, D. J. Bartlett, H. Courtois [MNRAS 545:staf1960, arXiv:2502.00121]
- [11] “Testing the local supervoid solution to the Hubble tension with direct distance tracers”; **R. Stiskalek**, H. Desmond, I. Banik [MNRAS 543:1556, arXiv:2506.10518]
- [12] “Symmetry in Hyper Suprime-Cam galaxy spin directions”; **R. Stiskalek**, H. Desmond [Res. Notes AAS 8 281, arXiv:2410.18884]
- [13] “Evaluating the variance of individual halo properties in constrained cosmological simulations”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz [MNRAS 534:3120, arXiv:2310.20672]
- [14] “Probing general relativistic spin-orbit coupling with gravitational waves from hierarchical triple systems”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [MNRAS 535:L1, arXiv:2307.01903]
- [15] “On the fundamentality of the radial acceleration relation for late-type galaxy dynamics”; **R. Stiskalek**, H. Desmond [MNRAS 525:6130, arXiv:2305.19978]
- [16] “Frequency- and polarization-dependent lensing of gravitational waves in strong gravitational fields”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [Phys. Rev. D 109, 124045, arXiv:2209.06459]
- [17] “The scatter in the galaxy–halo connection: a machine learning analysis”; **R. Stiskalek**, D. J. Bartlett, H. Desmond, D. Anbajagane [MNRAS 514:4026, arXiv:2202.14006]

- [18] “The dependence of subhalo abundance matching on galaxy photometry and selection criteria”; **R. Stiskalek**, H. Desmond, T. Holvey, M. G. Jones [[MNRAS 506:3205](#), [arXiv:2101.02765](#)]
- [19] “Are stellar-mass binary black hole mergers isotropically distributed?”; **R. Stiskalek**, J. Veitch, C. Messenger [[MNRAS 501:970](#), [arXiv:2003.02919](#)]

#### Co-Author – Submitted

- [1] “Forward-modelling the Tolman and distance-duality tests with IllustrisTNG”; H. Desmond, T. Yasin, **R. Stiskalek**, S. von Hausegger [[arXiv:2606.27182](#)]
- [2] “Learning the Universe: Constrained simulations of the Coma galaxy cluster – I. Radial X-ray and Compton- $y$  signatures”; U. P. Steinwandel, S. McAlpine, **R. Stiskalek**, Rüdiger Pakmor, V. Springel, E. Churazov, I. Khabibullin, J. Jasche, G. Lavaux, G. L. Bryan [[arXiv:2606.10028](#)]
- [3] “Testing cosmic anisotropy with cluster scaling relations”; T. Yasin, **R. Stiskalek**, H. Desmond, S. von Hausegger, P. G. Ferreira [[arXiv:2602.06007](#)]
- [4] “MEGATRON: how the first stars create an iron metallicity plateau in the smallest dwarf galaxies”; M. Rey, . . . , **R. Stiskalek**, . . . [[arXiv:2510.05232](#)]

#### Co-Author – Accepted

- [5] “MEGATRON: Reproducing the Diversity of High-Redshift Galaxy Spectra with Cosmological Radiation Hydrodynamics Simulations”; H. Katz, . . . , **R. Stiskalek**, . . . [[OJAp](#), in press, [arXiv:2510.05201](#)]
- [6] “MEGATRON: the impact of non-equilibrium effects and local radiation fields on the circumgalactic medium at cosmic noon”; C. Cadiou, . . . , **R. Stiskalek**, . . . [[OJAp 9:169640](#), [arXiv:2510.05667](#)]
- [7] “Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation”; V. Pandya, . . . , **R. Stiskalek**, . . . [[ApJ](#), in press, [arXiv:2604.06318](#)]
- [8] “Parameterizing Dark Energy at the density level: A two-parameter alternative to CPL”; G. Montefalcone, **R. Stiskalek** [[PRD 114:043538](#), [arXiv:2603.25735](#)]
- [9] “Testing subhalo abundance matching with galaxy kinematics”; F. Boreiko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [[MNRAS](#), in press, [arXiv:2601.07799](#)]
- [10] “Constraining cosmological simulations with peculiar velocities: a forward-modeling approach”; A. Valade, N. Libeskind, D. Pomarède, **R. Stiskalek**, Y. Hoffman, S. Gottlöber, R. B. Tully [[A&A](#), in press, [arXiv:2602.03699](#)]
- [11] “The subtle statistics of the distance ladder: On the distance prior and selection effects”; H. Desmond, **R. Stiskalek**, J. A. Najera, I. Banik [[MNRAS:stag1144](#), [arXiv:2511.03394](#)]
- [12] “The galaxy-environment connection revealed by constrained simulations”; C. Gallagher, T. Yasin, **R. Stiskalek**, H. Desmond, M. J. Jarvis [[MNRAS 546:stag108](#), [arXiv:2503.14732](#)]
- [13] “Renzo’s rule revisited: A statistical study of galaxies’ baryon–dark matter coupling”; E. Ko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [[MNRAS 544:4288](#), [arXiv:2508.03569](#)]
- [14] “CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning”; N. Huang, **R. Stiskalek**, J.-Y. Lee, A. E. Bayer, C. C. Margossian, C. K. Jespersen, L. A. Perez, L. K. Saul, F. Villaescusa-Navarro [[NeurIPS 2025](#), [arXiv:2507.03707](#)]
- [15] “The Manticore Project I: a digital twin of our cosmic neighbourhood from Bayesian field-level analysis”; S. McAlpine, J. Jasche, M. Ata, G. Lavaux, **R. Stiskalek**, C. S. Frenk, A. Jenkins [[MNRAS 540:716](#), [arXiv:2505.10682](#)]
- [16] “The CosmoVerse White Paper: Addressing observational tensions in cosmology with

systematics and fundamental physics”; Eleonora Di Valentino, . . . , **R. Stiskalek**,  
. . . [Physics of the Dark Universe, Volume 49, 2025, 101965, arXiv:2504.01669]

- [17] “Inferring the Ionizing Photon Contributions of High-Redshift Galaxies to Reionization with JWST NIRCам Photometry”; N. Choustikov, **R. Stiskalek**, A. Saxena, H. Katz, J. Devriendt, A. Slyz [MNRAS 537:2273, arXiv:2405.09720]

|               |                                                                                  |             |
|---------------|----------------------------------------------------------------------------------|-------------|
| INVITED TALKS | <i>Supernova-free local ladders for <math>H_0</math></i>                         | <b>2026</b> |
|               | Hubble & BAO Tensions, ICG Portsmouth                                            |             |
|               | GDR CoPhy Episode 4, Clermont-Ferrand, France                                    |             |
|               | Cosmology seminar, UCL                                                           |             |
|               | Cosmology Journal Club, University of Oxford                                     |             |
|               | <i>Bridging Machine Learning and Cosmology at Scale</i>                          |             |
|               | PV Reconstruction online meeting                                                 |             |
|               | <i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i> |             |
|               | SBI for Galaxy Evolution, Cambridge                                              |             |
|               | CAMELS + Baryon Pasting Workshop, University of Osaka, Japan                     |             |
|               | <i>Velocity Field Olympics</i>                                                   |             |
|               | Leibniz Institute for Astrophysics Potsdam, Germany                              |             |
|               | <i>Forward modelling of peculiar velocities and large-scale structure</i>        | <b>2025</b> |
|               | Institut d’Astrophysique de Paris                                                |             |
|               | <i>Pinning down the local Universe with digital twins</i>                        |             |
|               | University of Cambridge                                                          |             |
|               | Yale University                                                                  |             |
|               | <i>Do SH0ES Cepheids prefer the Planck Hubble constant?</i>                      |             |
|               | National Astronomy Meeting, Durham                                               |             |
|               | <i>Towards accurate models and tests of the local Universe</i>                   |             |
|               | Columbia University                                                              |             |
|               | Princeton University                                                             |             |
|               | Center for Computational Astrophysics, Flatiron Institute                        |             |
|               | <i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i> |             |
|               | Center for Computational Astrophysics, Flatiron Institute                        |             |
|               | <i>Velocity Field Olympics</i>                                                   |             |
|               | Center for Computational Astrophysics, Flatiron Institute                        |             |
|               | Cosmic Flows 2025, Brisbane                                                      |             |
|               | <i>Velocity Field Olympics</i>                                                   | <b>2024</b> |
|               | University of Portsmouth                                                         |             |
|               | <i>Search for the optimal dark matter halo density profile</i>                   | <b>2023</b> |
|               | University of Oxford                                                             |             |
|               | <i>Is the radial acceleration relation a fundamental correlation?</i>            |             |
|               | University of Oxford                                                             |             |
|               | <i>Frequency and polarization dependent propagation of gravitational waves</i>   | <b>2022</b> |
|               | University of Glasgow                                                            |             |
|               | LMU Munich                                                                       |             |
|               | Max Planck Institute for Gravitational Physics, Potsdam                          |             |
|               | <i>The scatter in the galaxy–halo connection</i>                                 |             |
|               | Baryon Pasters Collaboration meeting                                             |             |
|               | <i>The scatter in the galaxy–halo connection</i>                                 | <b>2021</b> |
|               | LMU Munich                                                                       |             |
|               | <i>Reversible-jump MCMC in gravitational-wave astronomy</i>                      | <b>2020</b> |
|               | Max Planck Institute for Gravitational Physics, Hannover                         |             |
|               | <i>Are binary-black hole mergers isotropically distributed?</i>                  |             |
|               | LIGO Scientific Collaboration Data Analysis telecon                              |             |
|               | <i>The relation between galaxies and dark matter halos</i>                       | <b>2019</b> |
|               | University of Oxford                                                             |             |